Release Notes

CADP internal QA build release notes — generated from OpenProject. Not the in-app Release Notes feature used by tenants.

1. Release Overview

Field	Details
Product Name	**CADP Workspace**
Release Version	**Release 1.0.0 (22)**
Release Date	**October 15, 2026**
Release Type	**Feature & Bug Fix Release**
Prepared By	**Development Team**
Approved By	**Ganagavathy**
Environment	**QA**

2. Release Summary

This release of CADP Workspace Version 1.0.0 (22) delivers 2 bug fixes, 7 new features, and 15 enhancements across the platform. Highlights include IS_NULL() and IS_NOT_NULL(): test whether a field is blank, Seed the new `popover` component (properties section + palette), and Every release-note template resolves completely (presets + custom). It also includes API changes (see API Changes). Refer to the Known Issues section for current limitations.

3. New Features

Feature ID	Feature Name	Description	Impact
	IS_NULL() and IS_NOT_NULL(): test whether a field is blank	Two new comparison functions in the formula builder. `IS_NULL({field})` is true when the; field is blank; `IS_NOT_NULL({field})` is its exact opposite. Both work in a formula field, in a; condition, and inside a `.where()` filter on a record aggregate.	There was no way to ask "is this field empty?". COALESCE was being used for it and gets it; wrong: COALESCE only falls through `null`/`undefined`/`NaN`, so it RETURNS an empty string rather; than falling through to the fallback — `COALESCE({a}, "x")` is `""` when `{a}` is a cleared text; box. `IF(IS_NULL({a}), "x", {a})` is the correct form.
	Seed the new `popover` component (properties section + palette)	Added the `popover` component to the seeders so it appears in the builder palette; (Layout group) and its Properties panel renders the **"Popover Properties"** section.; `seedComponentProperties.js`: new entry `fieldType: 'popover'` with; `propertyGroups.basic: [{ sectionName: 'Popover Properties', fields: [] }]`. The frontend; `PopoverProperties` panel reads `componentData` directly and writes every setting under; `basic.popover*`, so `fields:[]` is intentional (same as `modal` / `payment-mode`).; `seedComponentListHardcoded.js`: added `{ type: 'popover', label: 'Popover', icon:; 'MessageSquare', isAtomicComponent: false }` to the Layout Components group, after `modal`.	A popover is a trigger (icon, text or both) that opens a panel on the top, bottom, left; or right of it; the panel holds any dropped components. The palette and the properties sections; are DB-seeded, so a new component type needs both entries to show up.

Feature ID	Feature Name	Description	Impact
	Every release-note template resolves completely (presets + custom)	New `releaseTemplateFields.service.js`. Generate/Preview/Download load the selected template, and every; `{{token}}` resolves:; (a) document tokens;; (b) fact tokens derived from data: product_name, environment, build_number, previous_version,; release_owner/prepared_by/owner (plain workspace username; encrypted values are decrypted, never printed),; modules_affected, release_month, release_datetime, hotfix_title/version, change_counts, changelog_entries,; new_features_table, enhancements_table, bug_fixes_table, known_issues_table;; (c) any other token (preset fields such as deployment_steps/root_cause, or custom variables) is sent to the; AI by name as `templateFields`, falling back to "Not specified in the release information." The AI is told; never to invent IDs, numbers, dates, names or statuses.; Business Notes become structured `document.notes` (Highlights, Coming Soon, Important Notes, Migration Steps,; Known Issues with Description/Impact/Status, Limitations, Support). They are only expanded from what the author; wrote: an empty note stays empty, and the AI can no longer fill it from scratch.; `{{release_document}}` now includes those note sections, and the summary table has Category / Module / Item /; Status, including known issues, coming-soon and migration rows. Empty change sections read; "No bug fixes were documented for this release."; `draftRelease()` is the single drafting pipeline for Generate/Preview and generate-pdf-with-ai.; `normalizeFlowPositioning` unwraps lists whose items only hold block content, so a token placed as a bullet; (including in already-saved templates) doesn't render as one indented bullet.	
	IS_NULL() and IS_NOT_NULL() in the formula builder	The formula builder's Comparison group has two new entries, `IS_NULL()` and; `IS_NOT_NULL()`. `IS_NULL({field})` is true when the field is blank — empty, spaces only, or never; filled in. A `0` or an unticked checkbox is a value, not a blank. Autocomplete and the help panel; carry both, including the note that COALESCE differs.	There was no way to ask "is this field empty?" in a formula. COALESCE was being used for; it and returns an empty string instead of falling through, so `COALESCE({a}, "x")` is `""` for a; cleared text box. `IF(IS_NULL({a}), "x", {a})` is the correct form, and the help text says so.
	AI-failure banner; "Detailed Release Document" preset	PreviewTab shows an amber banner with the reason and a Retry button when the release's; AI integration was requested but failed (`aiStatus`). New Template Library preset; "Detailed Release Document" (category "Detailed Document"): header (version, title, date, type),; Key Highlights, `{{release_document}}` (numbered overview → module sections → summary table →; overall benefits → conclusion), then Additional Notes from Business Notes. Change-type presets; such as "Complete Release Outline" still group by New Features/Enhancements/…; the module-grouped; numbered layout needs `{{release_document}}`.	
	All 14 Template Library presets generate from real data	Presets no longer contain hardcoded sample content:; Feature-Based uses `{{module_sections}}`.; Change-Type uses new_features/improvements/bug_fixes/known_issues/breaking_changes.; Customer's placeholder bullets are replaced with real sections.; Version-/Date-Based use `{{change_counts}}` + `{{changelog_entries}}`.; Changelog uses `{{changelog_entries}}`.; Sprint uses list sections instead of `_1/_2/_3` fields.; Build's single-row tables are replaced with real `*_table` tokens.; Detailed Release Document is now header + `{{release_document}}`, which contains all notes.; `section()` renders a single-token section as a block (not a `<li>`).; PreviewTab:; unwraps block-only lists (mirrors the backend);; appends uncovered notes as rich HTML and skips them when the template uses `{{release_document}}`;; re-renders after a template switch so the new template's fields are drafted.; The editor's ReleaseNote variable list includes the new fact and table tokens.	
	Show/hide each part of the bar above the records list, from Transaction Settings	Transaction Settings has a new **Header & Toolbar** card (after Column Search). It lists; every part of the bar above a feature's records list, in screen order, each with a checkbox:; **Header:** Title · Published badge · Record count ("Showing 16 of 16 records").; **Toolbar:** Sort by · Sort direction (↑↓) · View buttons, with one row per view and an; *Opens in** picker (the default view) · Search box · Filters · Create.; ** ⋮ More actions menu:** Refresh · Import · Export · Settings (locked on) · Outbound API · Delete All.	

4. Enhancements / Improvements

Enhancement ID	Module	Description	Benefit
	transaction-parameters	new Access property **Record Lock Roles** (`recordLockRoles`, MULTI_SELECT of workspace; role ids, registered in `parameterMetadata.js`). When set, the Record Lock Field/Value lock only; holds callers who have one of those roles; other roles can still edit. Unset = everyone, exactly; as before. `recordAccessPolicy.js` adds `lockCoversCaller`; `evaluateRecordAccess` /; `evaluateRecordCreate` take `callerRoleIds`, and `assertRecordWritable` / `assertRecordCreatable`; read them from the request context, so neither save path (`featureData.service.updateFeatureData`,; `unified-event-processor` Submit) needed changing. `user.context.middleware.js` now puts; `roleIds` (token `role` + `workspaceRole`) into that context. A caller with no known roles; (worker, no request) is never held by a role-scoped lock. The workflow's Update/Upsert Record; nodes write to Mongo directly and never pass through this check.	RCF bugs 4/25 (PMT #1963): the Initiator could edit an RCF at every stage after submit.; A plain lock would also lock the Owner/COO, who must keep working on it. With Lock Field =; Workflow Phase, Values = VALIDATION, OWNER, ASSIGNEE, COO, ACTIONS_PENDING, CLOSED and Roles =; Initiator, only the Initiator is held, and only while the RCF is with someone else.
	release-notes	`/release/download-pdf/:id` (both the template and no-template branches) and; `/release/:id/generate-pdf-with-ai` no longer launch this service's Puppeteer. The HTML is; still built here (token substitution, flow fix, print-safe CSS); the PDF is rendered by; `cadp_template_backend_nodejs` `POST /templates/render/html-pdf` via a signed; `x-service-auth` token (same pattern as reportExport.service.js). Upstream failures map to 502.; `pdf-generation.service.js` remains only for SOP export.	The template service owns PDF generation; one Puppeteer pipeline.
	release-notes	The AI prompt (`buildAiPrompt`) now also asks for a structured `document`:; overview paragraphs → modules (grouped by each change's module) → per-change Title /; Description / Purpose / Illustrative Example (text, bullets, optional table) / Expected; Benefit / Note → overall benefits → conclusion. Existing response fields are unchanged.; New `releaseDocument.service.js` validates that JSON against the real confirmed items:; invented entityIds are dropped, skipped items get a deterministic block, all text is; HTML-escaped, and Module items aren't repeated as their own sub-block. Without an AI; integration, or when the AI fails, the same structure is built deterministically.; Generate/Preview responses now include `generatedJson.document` and `templateTokens`; (`release_document`, `module_sections`, `summary_table`, `overall_benefits`, `conclusion`,; plus rich `new_features`/`enhancements`/`improvements`/`bug_fixes`/`removed`/`deprecated`).; The no-template PDF/HTML uses `release_document`; template PDFs get the same tokens through; `buildTemplatePreviewData`. Content is still regenerated on every Generate/Preview/Download; (by decision — not persisted).	Generate should produce a proper release document, not a bullet list.
	release-notes	Generate/Preview responses now include `aiStatus { requested, used, error }`. A failed AI; call (e.g. the LLM gateway returning 502) is no longer silent: the error is mapped to a readable; message (gateway unavailable / credentials rejected / timeout / unreadable reply). Deterministic; fallback descriptions now read "The Owner Filter business rule has been added in this release."; instead of "Owner Filter (BusinessRule) has been new."	Generate showed basic content with no indication that the AI never ran (observed 2026-10-13: llmgatewaydev.ckdigital.in returned 502 for every request).
	release-notes	`resolveLlmGatewayUrl` (integration.service.js) sends chat requests to `<origin>/chat/completions`; when the integration's stored URL is the gateway origin or its model-listing endpoint (`…/models?…`).; Other stored paths are used as-is. The model-list and file-upload URLs already used only the origin and are unchanged.	The "AI Models" LLM integration stored `https://llmgatewaydev.ckdigital.in/models?...`, so every chat POST went; to /models and failed with 405 ("AI drafting failed — Request failed with status code 405" on release 07d9c13b…).; Verified live after the fix: release 07d9c13b… drafted by gpt-5.2 through that integration in about 10 seconds.
	release-notes	`confirmItems` now also accepts `generated` and `changes_requested` releases, so an author can go back to; Review Changes / Business Notes and run Generate again. Only `draft` advances to `items_confirmed`; other statuses; are kept, so a sent-back release stays in its approval stage. `in_review`, approved and published releases stay locked,; with a clearer error message.	Going back after Generate failed with "Cannot confirm items once a release has moved past review (status=generated)".
	release-notes	Generated release HTML (document blocks, notes, tables, changelog, delivered-items/grouped-changes) no longer; hardcodes font-size, font-family or color. It inherits from the element its token sits in, and headings and labels; differ only by weight.; `normalizeFlowPositioning` also strips `<hr>` and border-top/border-bottom lines outside tables, including in; already-saved templates. Table cell borders and boxed cards are kept.	In Generate/Preview, generated text rendered in a different size/style from the template text, and section; headings had horizontal lines.

Enhancement ID	Module	Description	Benefit
	release-notes	The template editor saves styles as `#id{…}` CSS rules, so the inline-only line strip missed them. `buildPrintSafeCss`; now appends an override (`hr` hidden; no border-top/bottom outside tables) after the template CSS. List unwrapping replaces; the list with a `div` that keeps its id/class, so the template's `#id` font-size/color still apply (generated text had; fallen back to a larger default). Single-item token-holder lists (e.g. `{{overview}}` as a bullet) are unwrapped too, so; all values line up; generated lists (inline list-style) are untouched.	
	workflow	`operationMatchesTrigger` (`workflow.event.mapper.js`) honours a new trigger config key; `runOnUpdate`. When it is `true` on a `feature-submit` trigger, the workflow also starts when an; existing record is submitted again (`operationType` `update`/`edit`). Unset or anything but; `true` keeps the 2026-10-08 behaviour (Feature Submit = create only, `d0c3116`); Feature Update; triggers are unchanged. The Workflow Builder's start-node panel shows the tick box for Feature; Submit events (frontend `StartNodeConfigPanel.tsx`); `workflow-nodes-guide.js` documents the key; for the AI builder.	RCF Tracker's "RCF Ticket Flow Manual" starts on the Submit RCF button and, by design,; restarts when the COO returns an RCF and the Initiator re-submits it (its first step only lets; a new or `INITIATOR`-phase RCF through). Since `d0c3116` that re-submit is an update, so the; trigger silently skipped it (`No matching trigger` is debug-level): RF00020 and RF00005 sat at; "Returned to Initiator" after the Initiator's Update saved fine.
	rcf-uat-defects	Transaction Settings → Access gets **Record Lock Roles** (`recordLockRoles`, a role; picker storing a JSON list of workspace role ids). With roles picked, the record lock holds only; users in those roles; unset = everyone, as before.; `recordAccessPolicy.ts`: `lockCoversCaller`, `useCallerRoleIds` (token `role` +; `workspaceRole`), `evaluateRecordAccess` takes `callerRoleIds`, new `useRecordAccessSettings`.; `useRecordAccessPolicy` now passes the caller's roles itself, so the View page (Edit button; hidden + form read-only when locked) works per role with **no change to `PreviewPage.tsx`**.; `TransactionManagement.tsx`: rows the caller may not edit get `canEdit: false`, so the row; menu (table/list/grid/card) shows no Edit — same rule as the View page. `KanbanCard.tsx` now; honours `record.canEdit` / `record.canDelete` like the row menu.; Settings screen: `role-picker` editor kind (`paramRowFreeze.ts`), `RolePicker`; (`ParamValueCell.tsx`), role names in `ParamValueDisplay.tsx`, shared `workspaceRoles.ts`; loader, catalog label in `useParamConfigRules.ts`, Access bucket mapping in; `transactionSettingsClosed.ts`.; The server enforces the same rule (master `recordAccessPolicy.js`, see its; `transaction-parameters.md`).	after submit the Initiator could Edit (and Delete) an RCF at every stage. Delete is a; role setting (Initiator → RCF Tracker → Delete = No). Edit needs "only on your turn": Lock Field =; Workflow Phase, Values = VALIDATION, OWNER, ASSIGNEE, COO, ACTIONS_PENDING, CLOSED, Roles =; Initiator. The Initiator keeps Edit before submit, when sent back (INITIATOR) and at Initiator; Review (REVIEW).
	release-notes	`GeneratedContent` gains `generatedJson.document` and `templateTokens`. PreviewTab; merges `templateTokens` into the custom-template token map, so `{{new_features}}` etc. render the; rich per-change blocks and the new `{{release_document}}`, `{{module_sections}}`, `{{summary_table}}`.; `{{overall_benefits}}` and `{{conclusion}}` tokens resolve. With no template, the fixed layout shows the; backend-rendered document, and the outline lists its modules, summary, benefits and conclusion.; The template editor's ReleaseNote variable list includes the new tokens.	
	release-notes	On the Business Notes tab the Generate button reads "Regenerate" once the release has been generated, and re-runs; confirm items → save notes → generate → Preview. Local status updates now mirror the backend (a regenerate or a Review; Changes save no longer resets the status).	
	release-notes	PreviewTab strips `<hr>` and border-top/bottom lines outside tables (mirrors the backend). Template Library; presets no longer draw lines under section headings or above footers.	
	release-notes	Scoped override CSS removes `hr` and border-top/bottom outside tables (including `#id` rules saved by the editor).; List unwrapping keeps the list's id/class so the template's font-size applies, and one-item token-holder lists are unwrapped.; Mirrors the backend.	
	release-notes	New `canEditRelease(status)` in releaseNoteDisplay.tsx. It is false for `in_review` (Pending Approval), `published`; and `archived`. The release list's row menu and the View page now hide Edit for those statuses (the list previously only; disabled it for published/archived, and the View page always showed it). The edit page goes read-only for the same; statuses, even when its URL is opened directly.	

5. Bug Fixes

Bug ID	Description
2024	BUG-01 — Old sessions stay listed as active long after the 60-minute session timeout
2025	BUG-02 — SSO Client Secret is sent to the browser in plain text in Edit Provider

6. Configuration Changes

Configuration	Description	Impact
None	No configuration changes are required for this release.	N/A

7. API Changes

API Name	Type	Description
Filter panel: negative operators replace the NOT toggle		Server search, export, extraction, lookup options and dashboard evaluate-table now understand the filter panel's new operators: `notEquals`, `notContains`, `notStartsWith`, `notEndsWith`, `greaterThanOrEqual`, `lessThanOrEqual`, `notBetween`, `isEmpty`, `isNotEmpty`. Each negative is evaluated as exactly "not the positive" (`$nor` in Mongo, `!` in memory), so it keeps the positive's multi-value (`a,b`) and numeric handling and matches records where the field is empty.
LLM integration as a template for form buttons (`llm-chat` with `inputs`)		`POST /integrations/llm-chat` accepts `{ integrationId, inputs }` (mapped name → form value) in addition to the existing `{ integrationId, model, content }`.; `integration.service.js`: `buildLlmTemplatePayload` (pure) treats the integration's saved `endpoints.payload` as the template — keeps `model`, extra params (temperature, max_tokens…) and system/assistant messages, and replaces the sample user message with the inputs as `Name:\nvalue` blocks; `{{Name}}` / `{{form}}` placeholders are filled instead when present. Inputs are made readable (rich-text HTML and the `{html, attachments}` envelope → text, lookup objects → display values, arrays joined, 12k chars max each). `executeLlmIntegrationTemplate` posts it with the stored key (Bearer), retries once without unsupported params, updates the integration's call counters and returns clear errors (400 no model / no form values, 504 timeout, 502 with the gateway's message).; `integration.controller.js`: `executeLLMChat` routes to the template mode when `inputs` is an object; the `content` mode is unchanged. Same route, so no RBAC/menu change.
Service-only HTML→PDF endpoint for the Release Notes export		Added `POST /templates/render/html-pdf` ({ html, css?, defaultFont?, paperSetup?, fileName? } →; application/pdf). It wraps the existing `pdfGenerationService.generatePdf` (no binding, no; pagination) and only accepts service tokens (`x-service-auth`, `req.user.kind === 'service'`);; browser tokens get 403. Called by cadp_client_ws_backend_nodejs for `/release/download-pdf`; and generate-pdf-with-ai. Verified over HTTP with a client-ws-signed service token.

8. Known Issues

Issue ID	Description	Workaround
	Role Management Change Effect	Because some menu structures have changed, certain features may return a 403 status code. To resolve this, edit the role, select the platform feature, and save the role again.
	Scheduler with Workflow Not Working	Use the conventional feature-wise approach wherever workflow scheduling is required.
	Formula Builder UI – Field Insertion Through Typing	This works in current fields. For external feature fields, it won't work, as there could be fields with same names from multiple features.
	Formula Builder – AI-Based Formula Writing/Editing Not Implemented	Use manual formula entry instead.
364	Organization Details → Basic Info: Website URL accepts invalid URL formats without validation.	Expected: Validate the URL format and show an error message for invalid values, such as www.example or example@123.www.example
	Builder Tool Not Available on Mobile Web	The builder tool is not supported on mobile web. This is implemented as expected — builder functionality is restricted to desktop browsers by design. It should be available on Tab though.
	Mobile – Formula Field Not Working in Tables & Charts	Formula fields are not evaluated correctly when used inside Tables and Charts on mobile. This is under review.

9. What to Test

- Login – Login, Sign In with Google (have to be configured in external providers in Tenants of metadata), Sign In with Microsoft (have to be configured in external providers in Tenants of metadata)

- **Create Application**
  1. Start from Scratch
  2. Choose/Create Vertical Template
- **AI-Assisted Creation & Ask FAQ** (Cavin's AI assistant — the AI launcher icon)
  1. **AI App Creation**
     a. From the workspace (any page other than Modules Management / Features Management / Builder Tool), open Cavin's AI and select the **Create** tab.
     b. Verify the helper line reads "Create builds a new application from your description".
     c. Describe the application to build and send. Verify creation progress is shown and "Application created from AI." appears on completion.
     d. Verify the generated application, its modules and features are listed and can be opened in the workspace.
     e. If the AI asks clarifying questions, answer them and verify creation continues (no success message is shown until something is actually created).
  2. **AI Module Creation**
     a. Open an application's **Modules Management** page, open Cavin's AI and select the **Create** tab.
     b. Verify the helper line reads "Create builds a new module here".
     c. Describe the module to build and send. Verify "Created N module(s) from AI." appears and the module list refreshes with the new module(s).
     d. Open the created module and verify it can be edited further.
  3. **AI Feature Creation**
     a. Open a module's **Features Management** page (or the Builder Tool), open Cavin's AI and select the **Create** tab.
     b. Verify the helper line reads "Create builds a new feature here".
     c. Describe the feature to build and send. Verify "Created N feature(s) from AI." appears and the feature list refreshes.
     d. From the Builder Tool, verify you are taken to the first created feature in the builder.
     e. Verify the generated form has appropriate components/fields, then publish it and verify records can be created, viewed and edited.
  4. **Create / Ask mode behaviour**
     a. In the Create tab, type a question instead of a build request. Verify nothing is created and an "Ask this in Ask mode" suggestion is offered; clicking it moves the question to the Ask tab.
     b. With a user who has only AI Builder or only AI Ask access, verify only that tab is available; with no create access, verify past conversations are read-only.
     c. Verify conversation history can be searched and reopened for each mode.
  5. **AI Ask FAQ**

     Scope: AI Ask FAQ is trained only on the following platform documentation. Questions must be asked on these topics only:

     - CADP User Login Guide
     - CADP Functional Specifications
     - CADP Erudite Management Complete
     - CADP Workspace Guide
     - CADP User Preferences Guide
     - CADP Role Management
     - AI FAQ Guide Transactions Runtime
     - CADP Login Complete
     - CADP Erudite Management Complete Guide
     - CADP Features Guide
     - CADP Modules Guide
     - CADP Platform Architecture Report
     - CADP Templates Integrations Workflows Basics Guide
     - CADP Features And Submodules Guide
     - CADP Organization Guide
     - CADP Builder Tool Guide

     a. Open Cavin's AI, select the **Ask** tab and ask a question about platform configuration on one of the trained topics above.
     b. Verify a relevant and accurate answer is returned.
     c. Ask a follow-up or ambiguous question and verify it is handled gracefully.
     d. Ask a question outside the trained topics and verify AI Ask FAQ declines or returns an appropriate out-of-scope fallback rather than a fabricated answer.
     e. Verify Ask mode changes nothing in the workspace (it only answers).

- **User Management**
  1. Create/Manage Users
  2. Create/Manage Groups
  3. Import/Export Users
  4. Archive/Unarchive Users
  5. Locked Users
  6. Waiting Approval
  7. Please update existing users in each workspace.
  8. In the Users tab, when creating a new user, the Approval Type default is No Approval.
- **User Preferences**
  1. Profile update – first name, last name updating, profile picture updating.
  2. Change password
  3. Locale and Date-time updating
  4. Interface
     i. Theme change, color change. Color change here will override the brand color for that specific user of that workspace only.
     ii. Records per page is reflecting.
     iii. Language (currently only English accepted)
  5. Customization – module and features are visible based on visibility setting, quick access for that enabled feature is redirecting to feature transactions (records) page.
  6. Privacy & Security
  7. Miscellaneous – Default landing page, preferred currency, auto-save drafts, save interval
  8. Preferred user workspace
  9. Reset Preferences
- **Groups**
  1. Archive/Unarchive

- **Roles**
    1. Platform Roles
    2. Archive/Unarchive
    3. Workspace Role
    4. Feature Access Level
    5. Data Access Level
    6. User or Group Role Mapping
- **Workspace Management**
    1. Create Workspace
    2. Clone Workspace
    3. Archive Workspace
    4. Workspace Settings
- **Organization Management**
    1. Organization Details
    2. Organization Address
    3. Branding Details
- **Module/Feature Management**
    1. Create/Manage Module
    2. Create/Manage Features
    3. Design components for features
    4. Features and Modules can be published
    5. Audit Logs (Platform & Workspace)
    6. Version Logs (Workspace & Workflow)
    7. Staging
    8. Feature Card and Module Card Customization
- **Notification & Alerts**
    1. Email Notification & Templates – Create template and use template to send notification
    2. SMS Notification & Templates – Create template and use template to send notification
    3. WhatsApp Notification & Templates – Create template and use template to send notification
    4. In-App Notification & Templates – Create template and use template to send notification
    5. Push Notification & Templates – Create template and use template to send notification

- **Erudite Management**
  1. **Configuration (prerequisite)**
     a. Enable Erudite Management in Role Management before testing.
     b. Verify users without the role cannot access the Erudite Management module.
  2. **FAQs**
     a. Create a new FAQ with a question, answer, and category (Account, Billing, General, Technical).
     b. Verify the FAQ appears in the FAQ list after saving.
     c. Enable "Display this on the dashboard" and verify the FAQ is visible on the dashboard.
     d. Edit an existing FAQ and verify changes are saved correctly.
     e. Delete a FAQ and verify it is removed from the list.
     f. Search for a FAQ by keyword and verify correct results are returned.
  3. **Documentation**
     a. Upload a document (PDF or DOC, max 5 MB) with a name, description, and category.
     b. Verify the document appears in the Documentation list after saving.
     c. Attempt to upload a file exceeding 5 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the document is visible on the dashboard.
     f. Edit document details and verify changes are saved correctly.
     g. Delete a document and verify it is removed from the list.
  4. **Tutorial Videos**
     a. Upload a video (MP4, MOV, or MKV, max 50 MB) with a name, description, and category.
     b. Verify the video appears in the Tutorial Videos list after saving.
     c. Attempt to upload a file exceeding 50 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the video is visible on the dashboard.
     f. Edit video details and verify changes are saved correctly.
     g. Delete a video and verify it is removed from the list.
  5. **End User Access**
     a. Verify end users can browse and read FAQs, documentation, and tutorial videos.
     b. Verify items marked "Display on dashboard" appear on the user's dashboard.
     c. Verify items not marked for dashboard display do not appear on the dashboard.

- **Components**
  1. Layouts
     1. Column
     2. Section (various nested scenarios)
  2. Data Management (API Integration, Device Integration)
  3. Input Components (Phone Number, Password, Dropdown with Feature as Field Value)
  4. Specialized Input & Display
  5. Display and Interaction
  6. Media and Files
  7. Forms and Validation
  8. Utility Components
  9. Data Config Component
  10. Hyper Link Component
  11. Favorite Component
  12. OCR Component
      a. Add the OCR component to a feature form and upload an image or document.
      b. General text extraction → verify the text is extracted from the uploaded image/document and displayed within the selected output textarea
      c. Verify the extracted text is saved correctly with the record and is visible on View/Edit.
  13. Analytics & Report Components
      1. Chart Component
      2. Gantt Chart Component
      3. KPI Card Component
  14. Form UI
      1. Table component and Table level formulas, Summary row formulas
  15. Reusable Component
- **Page Builder**
  1. Create a new page and verify it is saved and listed.
  2. Drag-and-drop layouts (columns, sections) and reusable components onto the page canvas.
  3. Configure component properties and verify changes reflect on the page.
  4. Reorder, move, and delete components on the canvas.
  5. Edit an existing page and verify changes are persisted.
  6. Delete a page and verify it is removed from the list.
  7. Publish the page and verify it renders correctly at runtime.
  8. Verify the page is responsive across mobile, tablet, and desktop.
  9. Verify role-based access — only authorized users can build/edit pages.
- **Workflow Management**
  1. Visual canvas for creating workflows
  2. Drag-and-drop node placement
  3. Node connection with visual links
  4. Canvas zoom and pan controls
  5. Grid snap for alignment
  6. Create new workflows
  7. Edit existing workflows
  8. Delete workflows
  9. List/view all workflows
- **Workflow Nodes – Control Flow**
  1. Start Node (workflow trigger point)
  2. End Node (workflow termination)
  3. Condition/Decision Node (if-else logic)
- **Workflow Nodes – Communication**
  1. Email Notification Node
  2. SMS Notification Node
  3. User Task/Approval Node
  4. System Notification Node

- **Workflow Monitoring**
  1. Execution history logs
  2. Workflow instance status tracking
  3. Node-level execution details
  4. Error logs and stack traces
  5. Approval node changes for user selection
  6. Added SLA and escalation reminder for approvals
  7. Added parallel workflow approval node
- **Printing Templates**
  1. QR Code – Generate and Print
  2. Barcode – Generate and Print
  3. Invoice – Generate and Print
- **Role-Based Dashboards**
  1. Role-based dashboard template creation
  2. Dashboard rendering based on created template
- **Scheduler & Task Scheduler**
  1. Create and use Schedule
  2. Create and use Task Schedule
  3. Event-based task schedule
  4. Schedule-based task schedule
  5. Schedule a notification
  6. SMS
  7. Mail
  8. WhatsApp
  9. Push
  10. In-App
  11. Schedule a report send (PDF) through mail from a configured email template.
- **Formula Builder Functionality**
  1. Same Feature
     a. Arithmetic
     b. Logical
     c. Comparison
     d. Text
     e. Math
     f. Date
     g. Conditional
     h. Miscellaneous
     i. Aggregate → with filter conditions and without filter conditions.
  2. Cross Feature
     a. Arithmetic
     b. Logical
     c. Comparison
     d. Text
     e. Math
     f. Date
     g. Conditional
     h. Miscellaneous
     i. Aggregate → with filter conditions and without filter conditions.
  3. Incompatible datatype operations, for example, arithmetic on a text input field, should show a validation error.
  4. Unrecognized fields, functions, or operators should show a validation error.
  5. If external fields are called without where() conditions, it should show MULTIPLE RECORDS FOUND ERROR in preview for non-aggregate functions/operations.
  6. Variables are not working. Review which variables should be displayed in the modal.

- **Formula Builder UI**
  1. On clicking a feature from the feature list, fields in the selected feature should be displayed, except for the current field, to prevent self-call loops.
  2. On selecting external feature fields, where() should appear in the modal.
  3. On selecting where(), the modal should automatically switch to the Feature and Fields tab.
  4. On hovering over a field, field information such as module name and feature name should be displayed.
  5. The Clear button or backspace should work as follows:
     i. First click of clear/backspace will select the token.
     ii. Next click of clear/backspace will clear that token.
  6. Empty formulas should not be saved.
  7. Formulas should not be saved without validation.
- **Formula Builder – Helper Tools**
  1. **Cheat Sheet** (**?** icon)
     a. Open the Cheat Sheet and verify functions are grouped by category (Lookup, Nested Lookup, Aggregation, Conditional, Math, String, Date).
     b. Search for a function and verify the list filters correctly.
     c. Click the copy icon on an example and verify it is copied to the clipboard.
     d. Press Esc and verify the modal closes.
  2. **Examples**
     a. Open the Examples drawer and verify 50 scenarios across all categories are listed.
     b. Filter by a category and search; verify results update.
     c. Click "Use This Pattern" and verify the formula is inserted into the editor.
  3. **Test Formula (Playground)**
     a. For a current-feature calculation (e.g. *{number_field} * 5*), open Test Formula, verify a type-aware input appears per field (number/date/true-false/text), enter values, run, and verify the correct result, type, and (where relevant) matched-record count.
     b. For a lookup formula (uses *.where()* / external feature fields), verify NO value-input form is shown and a "Lookup formula – test against a real record" message appears instead.
     c. For an aggregate formula (SUM/COUNT/AVG/MIN/MAX of records), verify the "Aggregate formula – test against a real record" message is shown and the form is not available.
     d. Verify the Test Formula button is disabled when the formula is empty.
  4. **Template Library**
     a. Open the Template Library and verify the **Most Used** tab lists the 20 platform built-in templates (e.g. GST, TDS, net salary, lookups, aggregations).
     b. Search and filter by category; verify results update.
     c. Click "Use This Template" and verify the formula is inserted into the editor.
     d. With a formula in the editor, click "Save current formula as template", fill title/category/description, save, and verify it appears under **My Templates**.
     e. Delete a saved (non built-in) template and verify it is removed. Verify built-in templates cannot be deleted.
  5. **Formula Wizard**
     a. **Lookup mode:** choose "Lookup a value", select a source feature, select the field to return, add a match condition (verify the "Right side is a field reference" option shows a dropdown of the current feature's fields), check the live preview, and Insert.
     b. **Aggregate mode:** choose "Aggregate records", select feature/field, optionally add one filter condition, choose the function (SUM/COUNT/AVG/MIN/MAX), and Insert.
     c. Verify the formula preview only appears after a return field is chosen, and that going Back keeps your previous selections.
     d. After inserting, hover over the tokens in the editor and verify the correct Module name and Feature name are shown for each (including the source feature for the looked-up field).
  6. **Smart Validation Messages**
     a. Enter an invalid formula (e.g. unknown function, missing where(), wrong argument count) and click Validate Expression; verify the error includes a helpful suggestion and an example, not just a generic message.
- **Integrations**
  1. Step 1: Go to the Integration Type field and select "Third-Party Software".
     Expected Result: A new dropdown field called Select Type should appear.
  2. Step 2: Check the options inside the Select Type dropdown.
     Expected Result: It should contain options including "Third-Party API" and "Google Drive".
  3. Step 3: Choose "Third-Party API" from the dropdown.
     Expected Result: The configuration fields specific to the Third-Party API should be displayed.
  4. Step 4: Change the selection to "Google Drive".
     Expected Result: The system should dynamically display the Google Drive configuration fields: Client ID, Client Secret, and Refresh Token.

This build — additional test focus:

- Approve/Reject `/execute` payload keyed by fieldId — `executeEventBinding` now builds `data` for `approve`, `reject` and `partial-approve` bindings with the same `buildSubmitStoragePayload` used by feature-submit: keys are field ids, `_metadata` is no longer sent, and `metadata.idTolabel` / `metadata.fieldCatalog` are always attached.; When no canvas field list is available, the form data is sent as-is (already fieldId-keyed) with `metadata.idTolabel` from the id→label map.; Extracted the shared record stamping (`currentDataVersion`, `__version`, `_createdByUsername`, `_operationType`, `saveFeatureDataPayload`, `id`) into one helper used by every branch.; Custom events and email/SMS triggers are unchanged (still label-keyed). Submit and Save as Draft are unchanged.

10. What to Test on Mobile

- Login – Normal Login
- Login – SSO (Sign In with Google, Sign In with Microsoft)
- Workspace Switch
- Module List
- Feature List
- Components
- Feature Data – Create, View, Edit, Clone, Delete
- Dashboard
- Integration – Weighing Machine, Third-Party API (Budgie, SAP), and Payment Gateways (Stripe, Razorpay, and PayU)
- Theme
- Push Notification
- Workflow – Approve and Reject
- Charts – Feature Data
- Data Table – Feature Data
- Security – Prevent copy, paste, cut, and text selection